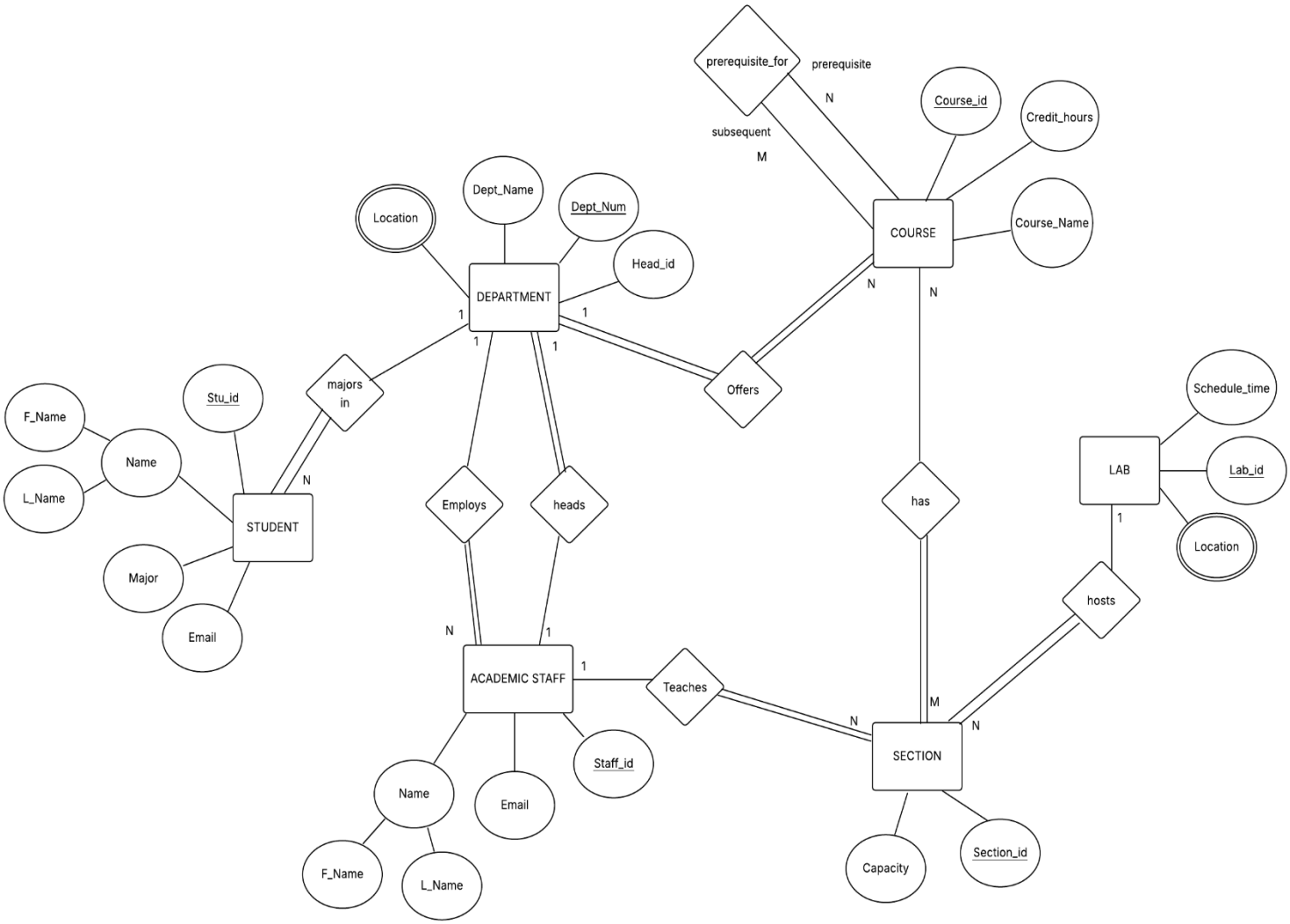
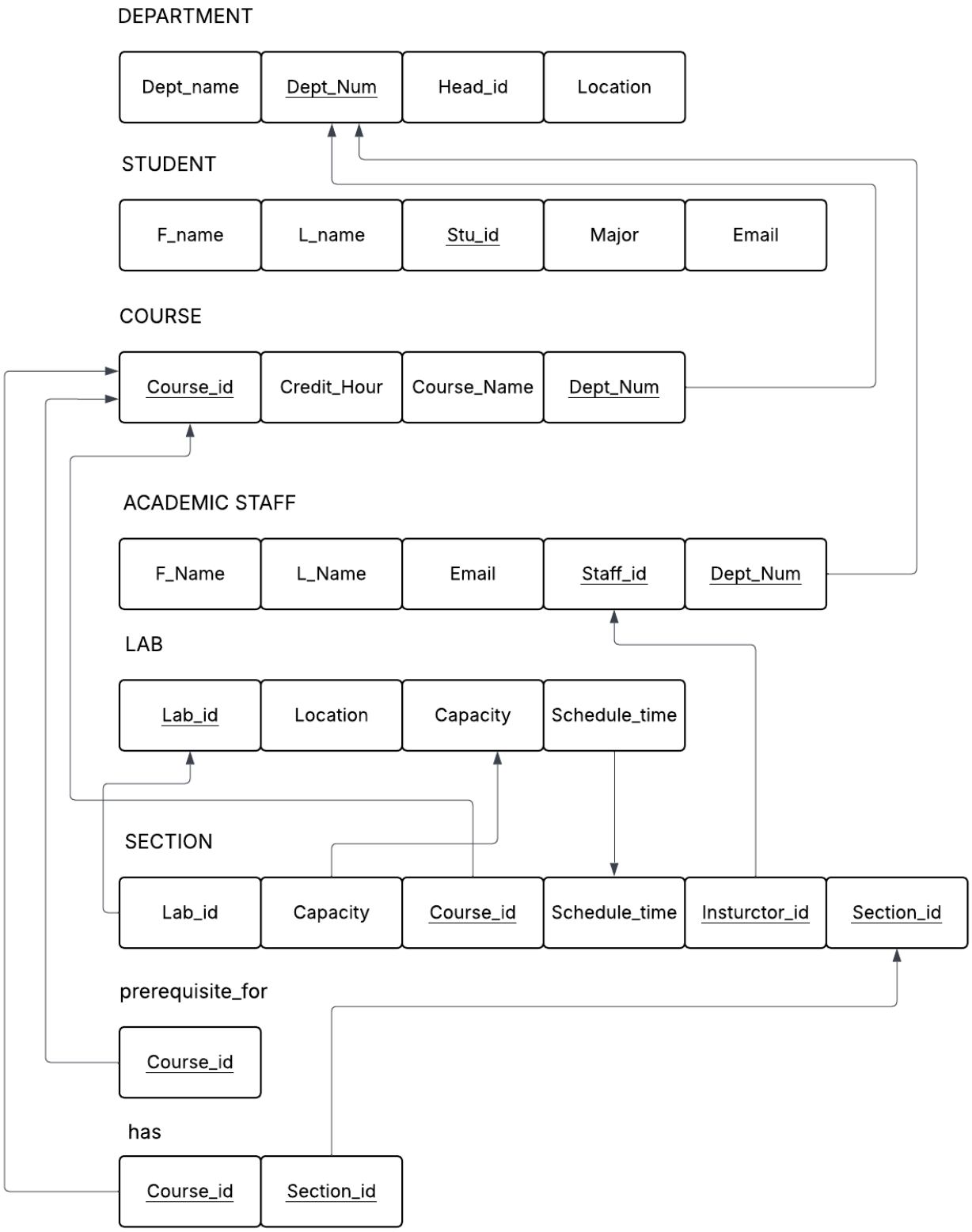
ER diagram


prerequisite_for
prerequisite
N
subsequent
M
Course_id
Credit_hours
Course_Name
COURSE
Location
Dept_Name
Dept_Num
Head_id
DEPARTMENT
1
1
1
1
N
N
Offers
majors in
Stu_id
F_Name
Name
L_Name
STUDENT
N
Major
Email
Employs
heads
has
Schedule_time
LAB
Lab_id
Location
1
hosts
N
1
ACADEMIC STAFF
1
Teaches
N
M
N
SECTION
Staff_id
Name
Email
F_Name
L_Name
Capacity
Section_id

Relational DataBase (RDB) diagram



Normalized relations

DEPARTMENT

Dept_name	Dept_Num	Head_id	Location

1NF:

DEPARTMENT

Dept_name	Dept_Num	Head_id

DEPARTMENT_LOCATION

Dept_Num	Location

2NF:

Table already in the second normal form

3NF:

Table already in the third normal form

LAB

Lab_id	Location	Capacity	Schedule_time

1NF:

LAB

Lab_id	Capacity	Schedule_time

LAB_LOCATION

Lab_id	Location

2NF:

Table already in the second normal form

3NF:

Table already in the third normal form

SECTION

Lab_id	Capacity	Course_id	Schedule_time	Insturctor_id	Section_id

1NF:

Table already in the first normal form

2NF:

Table already in the second normal form

3NF:

Table already in the third normal form

COURSE

Course_id	Credit_Hour	Course_Name	Dept_Num

1NF:

Table already in the first normal form

2NF:

Table already in the second normal form

3NF:

Table already in the third normal form

STUDENT

F_name	L_name	Stu_id	Major	Email

1NF:

Table already in the first normal form

2NF:

Table already in the second normal form

3NF:

Table already in the third normal form

ACADEMIC STAFF

F_Name	L_Name	Email	Staff_id	Dept_Num

1NF:

Table already in the first normal form

2NF:

Table already in the second normal form

3NF:

Table already in the third normal form

prerequisite_for

Course_id

1NF:

Table already in the first normal form

2NF:

Table already in the second normal form

3NF:

Table already in the third normal form

has

Course_id	Section_id

1NF:

Table already in the first normal form

2NF:

Table already in the second normal form

3NF:

Table already in the third normal form